

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Material name : ZEP CLEAN'EMS SPIRIT II DISINFECTANT
TOWELS_40CT_EA

Material number : 000000000000650800

Manufacturer or supplier's details

Company : Zep Inc.

Address : 11627 - 178 Street
Edmonton, Alberta T5S 1N6
Canada

Telephone : Compliance Services - 877-428-9937

Emergency telephone numbers

For SDS Information : Compliance Services - 877-428-9937

For a Medical Emergency : 877-541-2016 Toll Free - All Calls Recorded

**For a Transportation
Emergency** : CHEMTREC: 800-424-9300 - All Calls Recorded.

Recommended use of the chemical and restrictions on use

SECTION 2. HAZARDS IDENTIFICATION

Emergency Overview

Appearance	liquid
Colour	colourless
Odour	characteristic

GHS Classification

Eye irritation : Category 2A

Specific target organ toxicity - : Category 2
repeated exposure
(Inhalation)

GHS label elements

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

Hazard pictograms

:



Exclamation
mark

Signal word

:

Warning

Hazard statements

:

H319 Causes serious eye irritation.

Precautionary statements

:

Prevention:

P264 Wash skin thoroughly after handling.

P280 Wear eye protection/ face protection.

Response:

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P337 + P313 If eye irritation persists: Get medical advice/ attention.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture

:

Mixture

Hazardous components

Chemical name	CAS-No.	Concentration [%] *
2-(2-butoxyethoxy)ethanol	112-34-5	≥ 5 - < 10
tetrasodium ethylenediaminetetraacetate	64-02-8	≥ 1 - < 5

* The exact percentages of disclosed substances are withheld as trade secrets.

SECTION 4. FIRST AID MEASURES

General advice

:

Move out of dangerous area.

Show this safety data sheet to the doctor in attendance.

Do not leave the victim unattended.

If inhaled

:

If unconscious, place in recovery position and seek medical

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

	advice. If symptoms persist, call a physician.
In case of skin contact	: Wash off immediately with plenty of water for at least 15 minutes. If on skin, rinse well with water. If on clothes, remove clothes.
In case of eye contact	: Small amounts splashed into eyes can cause irreversible tissue damage and blindness. If in eyes, rinse with water for 15 minutes. Continue rinsing eyes during transport to hospital. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	: Keep respiratory tract clear. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician. Take victim immediately to hospital.
Most important symptoms and effects, both acute and delayed	: Effects are immediate and delayed. Symptoms may include irritation, redness, pain, and rash. Causes serious eye irritation. Review section 2 of SDS to see all potential hazards.
Notes to physician	: Treat symptomatically. Symptoms may be delayed.

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media	: Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.
Unsuitable extinguishing media	: High volume water jet
Hazardous combustion products	: Carbon dioxide (CO ₂) Carbon monoxide Smoke Nitrogen oxides (NO _x)
Specific extinguishing methods	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Further information	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Special protective equipment for firefighters	: Wear self-contained breathing apparatus for firefighting if necessary.

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

SECTION 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
- Environmental precautions : Prevent further leakage or spillage if safe to do so.
Prevent product from entering drains.
If the product contaminates rivers and lakes or drains, inform respective authorities.
- Methods and materials for containment and cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal.

SECTION 7. HANDLING AND STORAGE

- Advice on safe handling : Do not breathe vapours or spray mist.
Avoid contact with eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations.
- Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Electrical installations / working materials must comply with the technological safety standards.
- Materials to avoid : Store and keep away from, oxidizing agents and acids.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
2-(2-butoxyethoxy)ethanol	112-34-5	TWA (Inhalable fraction and vapor)	10 ppm	ACGIH

- Engineering measures : effective ventilation in all processing areas

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

Personal protective equipment

Respiratory protection : No personal respiratory protective equipment normally required.

Follow personal protective equipment guidance noted on the product label when using this product, especially if cleaning areas soiled with blood or body fluids.

Hand protection

Material : Protective gloves

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Safety glasses

Access to clean water to rinse eyes must be available, options include: eye wash stations or showers, or eye wash bottles with pure water.

Skin and body protection : For general disinfecting, no special protection is required. Follow personal protective equipment guidance noted on the product label when using this product, especially if cleaning areas soiled with blood or body fluids.

Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.

Hygiene measures : When using do not eat or drink.
When using do not smoke.

Wash hands before breaks and at the end of workday.

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	: liquid
Colour	: colourless
Odour	: characteristic
Odour Threshold	: No data available
pH	: 11.5 - 12.5
Melting point/freezing point	: No data available
Boiling point	: > 100 °C
Flash point	: > 93.3 °C Method: closed cup
Upper explosion limit	: not determined
Lower explosion limit	: not determined
Vapour pressure	: No data available
Density	: 1 g/cm ³
Solubility(ies)	
Water solubility	: soluble
Solubility in other solvents	: not determined
Partition coefficient: n-octanol/water	: No data available
Auto-ignition temperature	: not determined
Decomposition temperature	: No data available
Viscosity	
Viscosity, kinematic	: 3.8 mm ² /s (20 °C)

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: Stable
Chemical stability	: Stable under normal conditions.
Possibility of hazardous	: No decomposition if stored and applied as directed.

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

reactions
Conditions to avoid : No data available

Incompatible materials : Oxidizing agents
Acids
Alkali metals

Hazardous decomposition products : Carbon oxides
Nitrogen oxides (NOx)
Ammonia

SECTION 11. TOXICOLOGICAL INFORMATION

Potential Health Effects

Aggravated Medical Condition : None known.

Symptoms of Overexposure : Effects are immediate and delayed.
Symptoms may include irritation, redness, pain, and rash.
Causes serious eye irritation.
Review section 2 of SDS to see all potential hazards.
Treat symptomatically. Symptoms may be delayed.

Carcinogenicity:

IARC

No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

ACGIH

No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by ACGIH.

Acute toxicity

Product:

Acute oral toxicity : Acute toxicity estimate : > 5,000 mg/kg
Method: Calculation method

Skin corrosion/irritation

Product:

Result: Mild skin irritation

Method: Third Party Data - Actual or Inferred

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

Serious eye damage/eye irritation

Product:

Result: Eye irritation

Method: Third Party Data - Actual or Inferred

Respiratory or skin sensitisation

No data available

Germ cell mutagenicity

No data available

Carcinogenicity

No data available

Reproductive toxicity

No data available

STOT - single exposure

No data available

STOT - repeated exposure

Components:

tetrasodium ethylenediaminetetraacetate:

Exposure routes: Inhalation

Assessment: May cause damage to organs through prolonged or repeated exposure.

Aspiration toxicity

No data available

Further information

Product:

Remarks: No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

Toxicity to fish :
Remarks: Toxic to aquatic organisms.
Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Toxicity to daphnia and other aquatic invertebrates :
Remarks: Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
Toxic to aquatic organisms.

Persistence and degradability

No data available

Bioaccumulative potential

Product:

Partition coefficient: n-octanol/water : Remarks: No data available

Components:

2-(2-butoxyethoxy)ethanol :

Partition coefficient: n-octanol/water : Pow: 1

Mobility in soil

No data available

Other adverse effects

No data available

Product:

Additional ecological information : No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : Do not dispose of waste into sewer.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Dispose of in accordance with local regulations.
Pesticide wastes are acutely hazardous. Improper disposal of excess pesticide, spray mixture or rinsate is a violation of Federal law. If these wastes cannot be disposed of by use according to label instructions, contact your State Pesticide or Provincial Environmental control agency, or the Hazardous Waste Representative at the nearest EPA Regional Office for

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

guidance.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.

SECTION 14. TRANSPORT INFORMATION

Transportation Regulation (TDG) / Règlement Pour Le Transport (TMD): (Canada):
NOT REGULATED AS DANGEROUS GOODS OR HAZARDOUS MATERIAL

Transportation Regulation / Règlement Pour Le Transport: IMDG (Vessel):
NOT REGULATED AS DANGEROUS GOODS OR HAZARDOUS MATERIAL

Transportation Regulation / Règlement Pour Le Transport: IATA (Cargo Air):
NOT REGULATED AS DANGEROUS GOODS OR HAZARDOUS MATERIAL

Transportation Regulation / Règlement Pour Le Transport: IATA (Passenger Air):
NOT REGULATED AS DANGEROUS GOODS OR HAZARDOUS MATERIAL

Transportation Regulation / Règlement Pour Le Transport: 49 CFR (USA):
NOT REGULATED AS DANGEROUS GOODS OR HAZARDOUS MATERIAL

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

The product as delivered to the customer conforms to packaging requirements for shipment by road under Transport Dangerous Goods (TDG) Canada regulations. Additional transportation classifications noted above are for reference only, and not a certification or warranty of the suitability of the packaging for shipment under these alternative transport regulations.

SECTION 15. REGULATORY INFORMATION

This product has been classified according to the hazard criteria of the HPR and the SDS contains all of the information required by the HPR.

The components of this product are reported in the following inventories:

DSL All components of this product are on the Canadian DSL
TSCA On TSCA Inventory

For information on the country notification status for other regions please contact the manufacturer's regulatory group.

Inventory Acronym and Validity Area Legend:

TSCA (USA), DSL (Canada), NDSL (Canada)

SECTION 16. OTHER INFORMATION

WHMIS - GHS Label Information:

Hazard pictograms :



Exclamation
mark

Signal word :

Warning:

Hazard statements :

Causes serious eye irritation.

Precautionary statements :

Prevention: Wash skin thoroughly after handling. Wear eye protection/ face protection.

SAFETY DATA SHEET

ZEP CLEAN'EMS SPIRIT II DISINFECTANT TOWELS_40CT_EA

Version 1.2

Revision Date 12/21/2025

Print Date 12/21/2025

Response: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention.

Version:	1.2
Revision Date:	12/21/2025
Print Date:	12/21/2025

We believe the statements, technical information and recommendations contained herein are reliable, but they are given without warranty or guarantee of any kind. The information in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. Users should make their own investigations to determine the suitability and applicability of the information for their particular purposes. This SDS has been prepared by the Compliance Services organization supporting this manufacturer, supplier or distributor.